

Motion Control of Humanoid Robots Based on Combined Gait Planning and Reinforcement Learning

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Abstract: Humanoid locomotion control is an important research problem in robotics. Model-based methods can accurately characterize system dynamics, but they are sensitive to model inaccuracies and environmental disturbances. In contrast, reinforcement learning (RL) approaches exhibit strong robustness and adaptability, yet often suffer from low training efficiency, unstable convergence, and insufficient motion naturalness. In this paper, we propose a hybrid control framework that integrates human motion priors with reinforcement learning optimization. Within a motion imitation learning paradigm, human motion data are analyzed to construct a phase-driven, continuously adjustable reference gait. A trajectory-tracking reward based on the reference motion is then designed to guide policy optimization in a physics simulation environment. This enables the robot to preserve human-like motion structures while enhancing robustness to disturbances and environmental uncertainties. The proposed method is validated on the CASBOT-02 humanoid robot in both Isaac Gym and MuJoCo simulation environments, achieving stable locomotion control. Experimental results demonstrate that the approach effectively combines the advantages of motion imitation and reinforcement learning, significantly improving policy stability and generalization while maintaining motion naturalness.

Key Words: Gait Planning, Reinforcement Learning, Isaac Gym

1 Introduction

In recent years, humanoid robots have advanced rapidly. A wide variety of such robots have moved beyond the confines of laboratories and are frequently visible to the public. Promotions featuring humanoid robots showcasing their diverse human-like behaviors can be seen on social media platforms, at conferences and forums, and during marketing events. Currently, humanoid robots are gradually integrating into people's work and daily lives [1], functioning as human assistants. The purpose of designing humanoid robots is to enable them to replace humans in task execution, thereby freeing humans from repetitive, tedious, hazardous, and other similar work, and allowing people to engage in more free and creative work and life activities. In such work scenarios, it can be concluded that human walking and manipulation behaviors are of great importance. Thus, this paper focuses on researching human walking behaviors, aiming to enable humanoid robots to achieve human-like locomotion capabilities [4].

From the analysis of human walking behaviors [2], it can be observed that human walking primarily relies on coordinated movements of the foot joints, visual perception, and plantar tactile sensation. Specifically, for visually impaired individuals (blind people), who lack visual input, their walking primarily depends on plantar tactile sensation. By extension, they use white canes to expand the range of tactile perception. Another critical point is that, excluding the consideration of obstacles, the scenarios of human-environment interaction during walking [9] mainly include

flat ground, uneven terrain, slopes, and stairs. These types of terrain cover most of the daily walking scenarios of humans, and different terrains impose distinct requirements on body posture. When walking on flat ground, the primary considerations are the periodicity and regularity of gaits [17]. When walking on slopes, attention must be paid to the body's center of mass [3]; the projection of the center of mass on the ground should be maintained within the support area of the soles. When walking on uneven terrain [7], in addition to keeping the periodicity of gaits, it is necessary to cope with external force disturbances caused by the convexities and concavities of the terrain underfoot; that is, a certain level of anti-disturbance capability is required. For stair terrain, the core points are the foot-lifting height and the landing position; the landing position must be within a reasonable range on each step, otherwise, it is easy to fall.

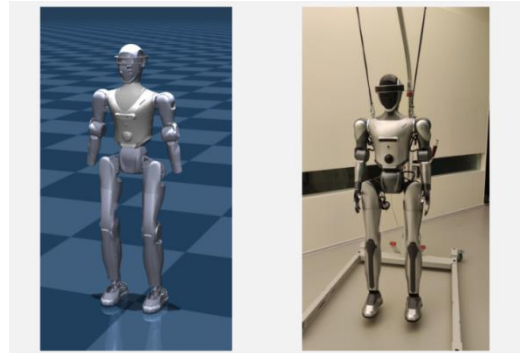


Fig. 1: CASBOT 02 Robot

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Currently, numerous studies have been conducted on the locomotion of legged robots across multi-terrain environments. Traditional approaches include several methods, such as Zero Moment Point (ZMP), Central Pattern Generator (CPG), Model Predictive Control (MPC), and Trajectory Optimization (TO). These model-based traditional methods offer high interpretability, mature theory, and stability; however, they struggle to adapt to complex terrains and lack adaptability and generalization. With the development of artificial intelligence and the maturity of algorithms such as neural networks, deep learning, and reinforcement learning, learning-based methods have been increasingly applied to robot motion control. Endowed with strengths such as strong adaptability, excellent generalization, and the ability to process high-dimensional perceptual information, learning-based methods enable robots to achieve more stable locomotion on complex terrains.

2 Relative Work

Regarding traditional methods, the ZMP [14] is an indicator for measuring the balance state of robots. By planning the robot's center of mass, the ZMP remains within the support area, ensuring the robot maintains balance. The CPG [21] aims to generate periodic joint angles or torques, which serve as reference trajectories for walking gaits. The periodic gaits generated by this method are more natural and smooth. MPC [20] establishes the robot's dynamic model and converts the control problem into a nonlinear optimization problem. It takes the robot's hardware constraints, posture, dynamic equations, contact states, and other factors as constraints. Under the condition that these constraints are satisfied, it achieves the artificially set optimization objectives and reaches a local optimum at each moment. TO [18] is similar to MPC in that it reformulates the control problem as a nonlinear optimization problem. The difference from MPC is that TO aims to obtain a globally optimal trajectory.

In terms of learning-based methods, a growing number of relevant studies have been conducted in recent years, with research directions generally falling into three categories: reinforcement learning (RL)-based methods, hybrid methods combining models with RL, and hybrid methods combining imitation learning with RL. For methods relying entirely on RL [8, 11], the robot learns standard walking postures solely through the design of reward functions. This enables the robot to possess stronger generalization; however, such methods have obvious drawbacks, including the cumbersome design of reward functions and slow convergence during learning and training. The latter two categories are similar, differing primarily in how the policy model acquires reference gaits. For model-based hybrid methods [5, 12], the robot's kinematic and dynamic models are established, and reference gaits that are well-suited to the robot's motion state are directly generated via constraint optimization. For imitation learning-based hybrid methods [6, 16], the robot's reference gaits are derived from human motion imitation as demonstration data. Due to differences in body structure and motion capabilities between robots and humans, human demonstration data typically require motion retargeting [19]; only after retargeting can the data be used for robot training.

This study builds upon recent advances in the literature and conducts simulation-based validation on the CASBOT-02 humanoid robot, as shown in Fig. 1. A human motion imitation reinforcement learning framework is adopted. By analyzing human walking characteristics together with the structural properties of the robot, a phase-driven, parameterized joint-angle function generator is constructed to generate reference gait trajectories. In combination with a trajectory-tracking reward, the control policy is optimized within the reinforcement learning framework, enabling the robot to achieve stable locomotion.

3 Method

First, regarding the hybrid method combining model-based control and reinforcement learning, we design a reference trajectory function generator that conforms to the robot's walking gaits through gait planning. The gait trajectories generated by this generator serve as observational information for the Critic in the policy network, enabling it to make accurate evaluations of actions. The Proximal Policy Optimization algorithm is adopted as the policy network optimization method, and the Isaac Gym [10] is used as the simulation environment for training.

3.1 Humanoid Robot and Reinforcement Learning

We model the robot's policy using the reinforcement learning framework $M = \langle S, A, T, O, R, \gamma \rangle$, where S and A denote the sets of state space and action space, respectively; $T(s'|s, a)$ represents the state transition dynamics, describing the probability of transitioning to state s' after executing action a in state s ; $R(s, a)$ is the reward function, measuring the immediate gain of executing action a in state s ; γ is the discount factor, used to balance the weights of immediate rewards and future rewards; and O denotes the observation space.

In the Isaac Gym simulation environment, a wealth of practical privileged environmental information can be leveraged to accelerate the policy model's convergence. However, such privileged information is unavailable in real-world environments. Therefore, the policy model is constructed as an asymmetric Actor-Critic network [22]. The observation space dimensions of the Critic and Actor networks are inconsistent. The overall optimization objective is to find an optimal policy π that maximizes the expected cumulative reward.

$$\mathbb{E}[R_t] = \mathbb{E}[\sum_t \gamma^t r_t] \quad (1)$$

We adopt the loss function of Proximal Policy Optimization (PPO) [13]. Derived from Trust Region Policy Optimization (TRPO), PPO offers a simpler algorithm, faster learning iterations, and quicker convergence. PPO is of the clipped type, which limits the range of change in the probability ratio between the new and old policies during policy updates. This ensures policy improvement while preventing excessive updates and enhancing training stability. The policy loss function is defined as follows:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(\hat{r}_t(\theta) \hat{A}_t, \text{clip}(\hat{r}_t(\theta), 1-\epsilon, 1+\epsilon) \hat{A}_t \right) \right] \quad (2)$$

Where $\hat{A}_t$ denotes the advantage function, and ϵ represents the clipping range.

The relevant configuration parameters for training the policy model are presented in Table 1 below. These parameters mainly include the number of parallel environments, the discount factor, the generalized advantage estimation discount factor, the learning rate, the activation function, and the hidden layers of the Actor and Critic networks.

Table 1: Model Configuration Parameter

Parameter	Value
Environment Number	10000
Entropy Coefficient	0.001
Learning Rate	1e-5
Discounting Coefficient	0.994
GAE lam	0.9
Activation Function	ELU(x)
Actor Hidden Dimension	[512, 256, 128]
Critic Hidden Dimension	[512, 256, 128]

3.2 Gait and Observation

The core of the reinforcement learning training process lies in optimizing the policy through reward signals to achieve predefined objectives, where the reward function is strongly correlated with the target task. To accelerate the convergence of the policy model, we incorporate a normalized gait planning scheme. Specifically, we design a foot joint angle function generator to emulate the periodic walking pattern of the robot. A smooth and natural cosine function is employed as the reference trajectory C_{foot} for the joint angles, primarily modeling the pitch motions of the hip, knee, and ankle joints. Within each gait cycle, each leg is divided into stance and swing phases. The gait period is set to 1 second, with the swing and stance phases each lasting 0.5 seconds. Accordingly, the general mathematical formulation can be expressed as follows:

$$C_{foot} = 0.5 \cdot a \cdot (\cos(2\pi \cdot c \cdot t) - 1) - b \quad (3)$$

The terrain constructed in the simulation environment for robot interaction primarily consists of flat ground. The objective of this study is to enable omnidirectional locomotion of the robot on flat terrain, including forward walking, backward walking, lateral movement, turning, and standing. The parameters a , b , and c are individually configured to accommodate different locomotion modes. Considering that walking speed also influences the gait trajectory, we design three reference gaits corresponding to forward motion, standing, and lateral–turning–backward motion, respectively. The velocity settings of the policy are summarized in the Table 2.

Table 2: Velocity Configuration

Category	Unit	Range
X Linear Velocity	m/s	[-0.3, 0.6]
Y Linear Velocity	m/s	[-0.3, 0.3]
Body Angle Velocity	Rad/s	[-0.3, 0.3]

In the forward mode, to ensure adaptability to different walking speeds, the reference gait is designed based on a nominal speed of 0.4 m/s. The reference trajectory for

standing is set to the default joint positions. For the remaining three locomotion modes, since their velocity variations are relatively small, we adopt an in-place stepping gait trajectory. Therefore, the reference gait trajectory mainly consists of the variations of the hip, knee, and ankle joints in the pitch direction, while the reference values in the other directions are set to zero. Although the reference gait trajectories are predefined and fixed, the mutual constraint between the gait tracking reward and the velocity tracking reward enables adaptive coordination between speed and gait. The reference gait configurations for each mode are presented in the Table 3. In the standing mode, the default joint angles are set to $[-0.18, 0.0, 0.0, 0.36, -0.18, 0.0]$, with the left and right legs configured symmetrically. In addition, the reference gait velocity is obtained as the time derivative of the corresponding position reference trajectory.

Table 3: Gait Configuration

Locomotion Mode	Swing phase/a-b-c	Stance phase/a-b-c
Forwarding-Hip	[0.24, 0.26, 2]	[-0.16, 0.26, 1]
Forwarding-Knee	[-0.7, -0.3, 2]	[-0.3, -0.3, 2]
Forwarding-Ankle	[0, 0.36, 0]	[0, 0, 0]
Other Mode-Hip	[0.32, 0.18, 2]	[0.0, 0.18, 2]
Other Mode-Knee	[-0.69, 0.36, 2]	[0.0, 0.36, 2]
Other Mode-Ankle	[0, 0.18, 0]	[0, 0.18, 0]

In the simulation environment, under the asymmetric Actor-Critic network structure, various types of effective privileged information can be fed into the Critic network. In contrast, to align with real-world environmental information, the observation input to the Actor network includes only proprioceptive information (IMU). The observation spaces of the entire policy network are presented in Table 4. Both the Actor and Critic networks are fed historical observation information, i.e., observations from the previous several time steps are input to the policy network together. This approach can compensate for partial observability and enable the policy network to recover environmental dynamic characteristics through temporal dependence, thereby making more reasonable action decisions.

Table 4: Observation Space

Observation	Actor	Critic
Commands(V_x , V_y , V_{yaw})	✓	✓
Joint Velocity	✓	✓
Joint Default Position Errors	✓	✓
Last Actions	✓	✓
Gait Reference	✓	✓
Base Orientation	✓	✓
Base Angle Velocity	✓	✓
Base Linear Velocity	×	✓
Environment Friction	×	✓
Environment Distribution	×	✓
Body Mass	×	✓
Contact Force Mask	×	✓

The actions output by the policy are converted into corresponding torques via a PD controller, which are then transmitted to the joint actuators of the robot in the simulation environment, as shown in Equation 3. The policy actions represent the target joint angles under the current state; the error between these target angles and the current joint angles is converted into corresponding joint torques through the proportional gain K_p . A damping term for joint velocity is added to the equation to suppress excessive joint velocities, with the damping coefficient denoted as K_d .

$$\tau = K_p (q_{\text{target}} - q) - K_d \dot{q} \quad (4)$$

Where q_{target} denotes the target joint position, q represents the current joint position.

3.3 Domain Randomization and Reward

When operating in real-world environments, humanoid robots are subject to various uncertainties that cannot be neglected, unlike the idealized conditions typically assumed in simulation [24].

Table 5: Domain Randomization

Category	Unit	Range
Environment Friction	-	[0.1, 2.0]
Environment Restitution	-	[0.0, 0.5]
Base Mass	kg	[-7.0, 7.0]
Link Mass(scale)	-	[0.8, 1.2]
Push Linear Velocity	m/s	[-0.3, 0.3]
Push Angle Velocity	Rad/s	[-0.4, 0.4]
Disturbance Force	N	[-500, 500]
Stiffness Gain(scale)	-	[0.8, 1.2]
Damping Gain(scale)	-	[0.8, 1.2]
Motor Strength(scale)	-	[0.7, 1.4]
Motor Offset	rad	[-0.035, 0.045]
Motor Delay	s	[0.05, 0.15]
CoM	m	[-0.06, 0.06]
Actions Delay	s	[0.05, 0.15]
IMU Delay	s	[0.01, 0.05]

On the one hand, the external environment and contact conditions exhibit significant variability. For example, the contact friction coefficient may vary across different regions of the same terrain, and the robot may experience unavoidable random collisions or external impacts during locomotion. On the other hand, uncertainties and time-varying characteristics also arise from the robot's own dynamics and actuator properties. As the robot is used over time, joint wear may occur, leading to changes in joint friction and damping parameters. The overall mass distribution of the robot may deviate due to payload variations or modeling errors, resulting in shifts in the center of mass, and the link mass parameters may differ from those assumed in simulation. In addition, controller parameters, such as the proportional gain (KP), may be subject to tuning inaccuracies in the real system. Regarding the actuators, motor output strength may fluctuate, static motor offsets may be present, and command execution is typically affected by communication and computation delays. These

factors collectively introduce non-negligible discrepancies between the real system and its simulation model. Therefore, to enhance locomotion robustness in real-world environments, it is necessary to model environmental disturbances, contact variations, and uncertainties in robot dynamics and actuator parameters within the simulation. Domain randomization is employed to inject disturbances and randomize system parameters during training. The specific types of perturbations are summarized in Table 5.

The robot's perceptual information is derived from sensors. As mentioned in the previous section, the robot's policy model executes actions based on IMU data as observations, including joint positions, joint velocities, and robot posture. In real-world scenarios, the data transmitted from sensors contains observation noise. Therefore, during training in the simulation environment, to enhance the policy model's robustness, it is necessary to add observation noise to the observations. This part of data processing is implemented in the observation space of the Actor network, with specific types of noise presented in Table 6.

Table 6: Domain Randomization

Category	Unit	Range
Joint Position	rad	[-0.05, 0.05]
Joint Velocity	Rad/s	[-0.5, 0.5]
Body Angle Velocity	Rad/s	[-0.1, 0.1]
Body Linear Velocity	m/s	[-0.05, 0.05]

The core of training the policy model via reinforcement learning lies in the design of the reward function. To enable the robot to walk stably, in addition to the gait-reference-trajectory reward mentioned earlier, we have set rewards for gait planning, speed command, body posture, and energy consumption. For gait planning, penalties for foot slippage and contact force have been added. Regarding movement commands, we have established rewards for forward progression and yaw. Meanwhile, while ensuring the robot can follow the movement commands, penalties are imposed on behaviors such as the robot's body bouncing or tilting over. For the body posture, a reward for maintaining the height of the robot's center of mass (CoM) is set. During walking, the height of the CoM basically remains within a stable range relative to the height of the lower body. At the same time, penalties are applied when the Euler angles of the robot body around the x-axis and y-axis are non-zero, as the presence of such angles implies that the robot may fall over. In the reward weight design, higher weights are assigned to the gait tracking and velocity tracking terms, while maintaining a moderate balance between them. This ensures mutual constraint between gait tracking and velocity tracking, thereby enabling the policy to achieve omnidirectional locomotion. In addition, a relatively larger weight is allocated to the posture stabilization term to prevent the robot from falling and to enhance overall stability.

The main reward components and their corresponding weights are presented in Table 7. Herein, explanations for the relevant symbols are provided as follows: $\text{Clip}(x, a, b)$ denotes confining the range of x within $[a, b]$; the symbols H, K, and A represent the hip joint, knee joint, and ankle joint respectively; p , r , and y denote orientations

corresponding to pitch, roll, and yaw respectively; L and R stand for left and right respectively; f, h, and G represent force, height, and gravity respectively. Besides, the following expressions are defined:

$$\phi(w, e, i) = \exp(-w \times \|e\|_i) \quad (5)$$

Table 7: Reward Design

Reward Name	Function	Weight
Joint Position	$\sum_{\theta(H, K, A)} \phi(5, \theta - \hat{\theta}, 2)$	3.0
Feet Joint Position	$\sum_{\theta(A)} \phi(8, \theta - \hat{\theta}, 2)$	0.8
Joint Default Position	$\phi(100, \theta_{Hr, Hy, Ar} - 0.1, 2)$	0.2
Joint Velocity	$\sum_{\theta(H, K, A)} \phi(0.3, v - \hat{v}, 2)$	0.5
Feet Slip	$(V_{Ax}^2 + V_{Ay}^2)^{\frac{1}{2}}$	-0.2
Feet Contact Force	$\sum_{f_L, f_R} \text{Clip}(f - f_{limit}, 0, 500)$	-0.001
Linear Velocity	$\phi(5, V_{xcmd, ycmd} - V_{x, y}, 2)$	1.5
Angle Velocity	$\phi(5, w_{yawcmd} - w_{yaw}, 2)$	1.8
Linear Velocity Mismatch	$\phi(10, v_z, 2)$	0.5
Angle Velocity Mismatch	$\phi(5, w_{p, r}, 2)$	0.5
Orientation	$\phi(10, \theta_{p, r}, 1) + \phi(20, G_{p, r}, 2)$	1.0
Base Height	$\phi(100, h - h_{target}, 1)$	0.2

4 Experiments

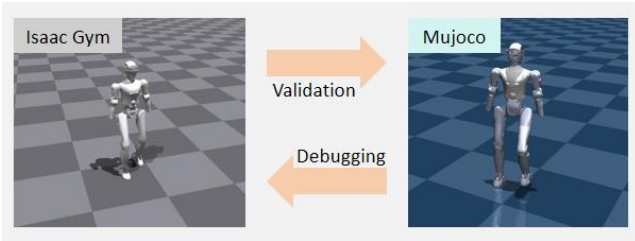


Fig. 2: Combination Simulation

The overall methodology of our experiment is as follows: the policy model is trained in the Isaac Gym simulation environment, and after obtaining stable locomotion gaits, the model is transferred to the constructed MuJoCo simulation environment [23] for validation. Isaac Gym enables large-scale parallel reinforcement learning training, significantly reducing model training time and offering substantial advantages. In contrast, MuJoCo is well-known for its accurate rigid-body dynamics and contact models, and its high-precision physical simulation environment is highly suitable for validating the model policy. The overall experimental framework is illustrated in Fig. 2.

4.1 Isaac Gym Simulation

As described in the previous section, to accelerate model training and improve robustness, gait planning and disturbance simulation were introduced, and corresponding reward functions were carefully designed. The results obtained in the Isaac Gym simulation environment are shown as follows. Specifically, Fig. 3 presents the gait tracking reward, Fig. 4 shows the velocity tracking reward, and Fig. 5 illustrates the robot posture reward. The convergence of these key rewards indicates that the robot’s motion in the simulation has gradually become stable and that gait tracking for five different motion modes has been successfully achieved.

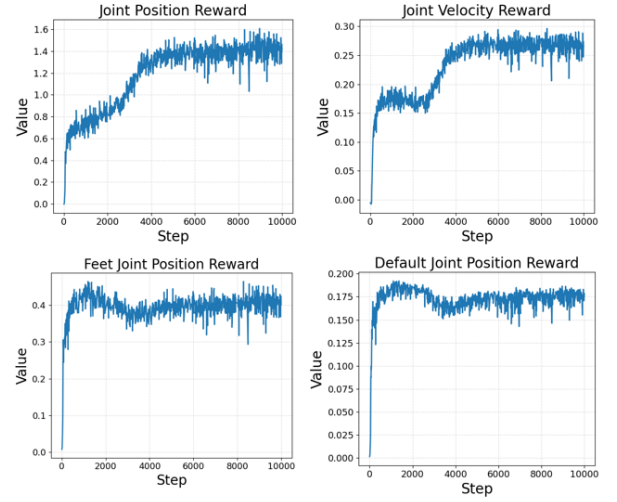


Fig. 3: Gait Tracking Reward

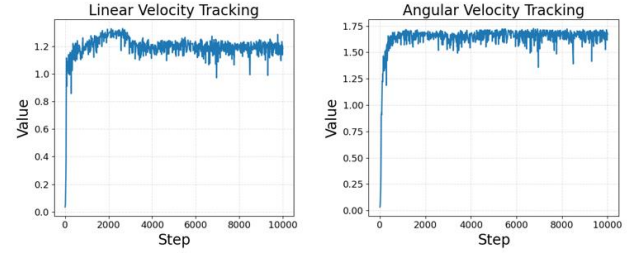


Fig. 4: Velocity Tracking Reward

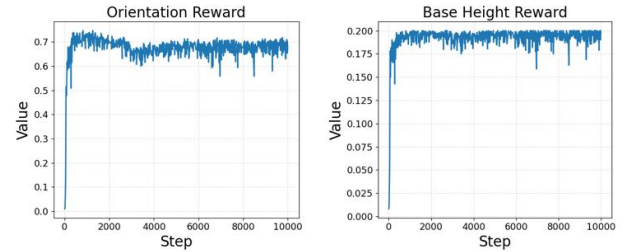


Fig. 5: Pose Reward

4.2 Mujoco Simulation

After obtaining a stable locomotion policy through training in the Isaac Gym environment, the learned policy was deployed in the MuJoCo simulation environment to validate the locomotion gait. From the real-time simulation and the corresponding gait tracking curves, it can be observed that the robot maintains stable locomotion. Fig. 6 illustrates the robot's motion process in the real-time simulation under five different motion modes, where A–E represent standing, lateral movement, turning, backward motion, and forward locomotion, respectively. Fig. 7 shows the reference gait tracking performance in the forward locomotion mode, specifically including the position and velocity curves of the hip, knee, and ankle joints in the pitch direction.

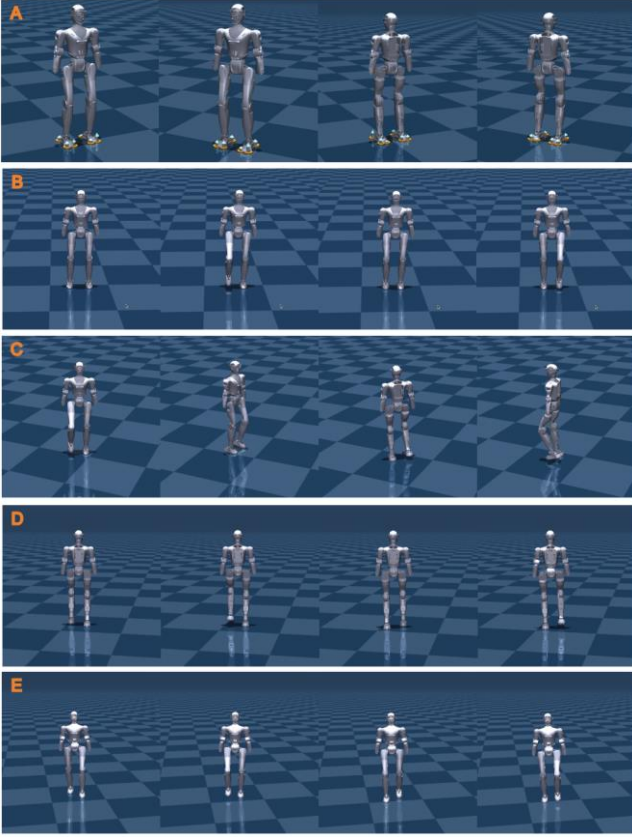


Fig. 6: Gait of Mujoco

5 Conclusions

By leveraging a motion imitation reinforcement learning framework, we integrate gait planning with reinforcement learning to enable stable locomotion of the robot. Simulation results demonstrate that the proposed method is feasible and effective. However, the current simulation environment only considers flat terrain. In future work, we will introduce various complex terrains to improve the robot's adaptability to challenging environments [15], such as rough surfaces, slopes, and stairs. Furthermore, diversified environmental disturbances under complex conditions will be incorporated to enhance the generalization capability of locomotion. We also plan to integrate visual perception information to improve gait adaptability across different terrains.

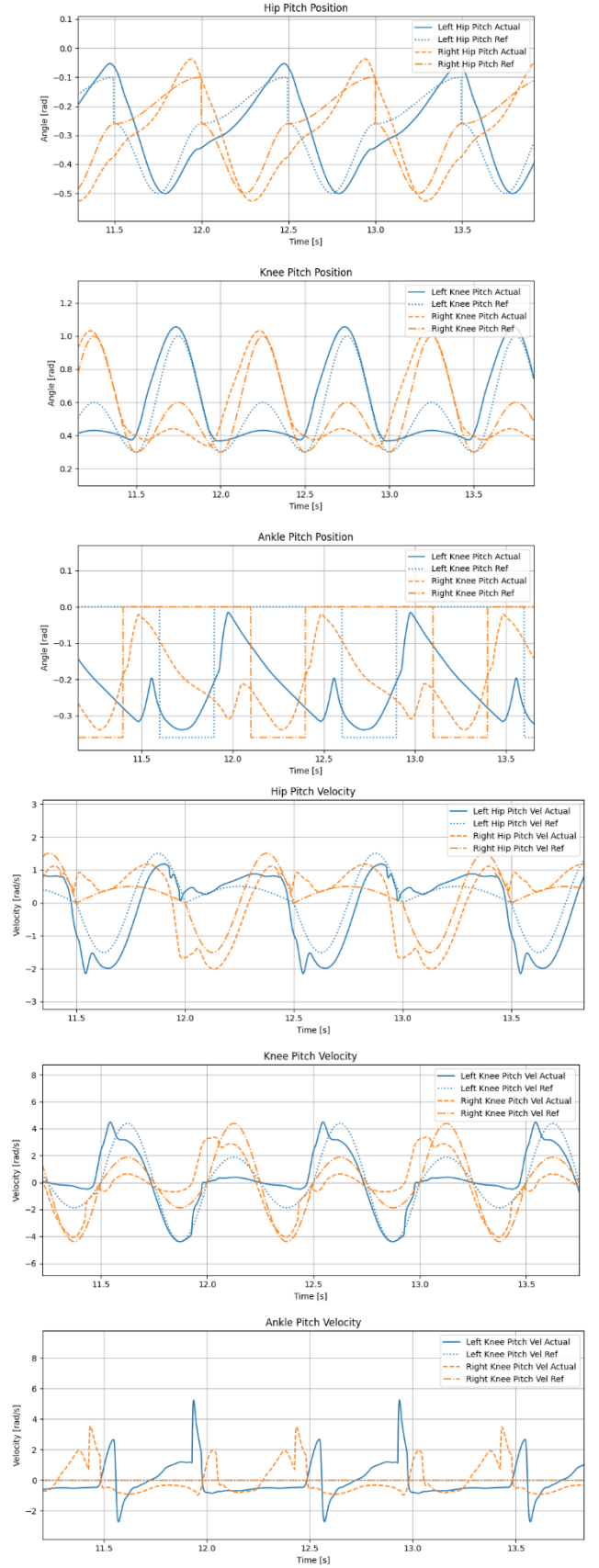


Fig. 7: Gait Trajectory In the Forward Mode

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